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Chapter 13
Artificial Intelligence Solutions for the Visually Impaired: A Review

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ABSTRACT

The major population of visually impaired and blind peoples were overlooked for years. Technology always keeps advancing and being developed in all aspects. Numerous solutions were being discovered for any current day problems of humans, but not for the people having low vision, partially impaired vision, and blind people. In this paper, the authors focused on the research papers that are available on the topic of AI solutions for the visually impair and reviewed the papers. This chapter is different from other review papers, giving the review of more than 30 research papers in one place, which are speaking about the new concepts that would make people's lives easy. The review paper covers the research paper, the technology used, the solutions offered, and their solutions. Some recommendations are also being given based on the limitations found in the different papers.

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INTRODUCTION

According to WHO, across the globe there are more than 285 million visually challenged people. And according to a report estimated No. of visual impaired people in India is more than 66 million; of these, 57 million persons have low vision, and 9 million are blind (Bagwan & Sankpal, 2015). The daily life of people having problems is far different than people with normal vision. Being blind or having someone close to us with low or blind vision needs effort and care towards people.

Eyesight is the most important sense a person needs to have for performing his task. Visually impaired or blind people needed most assistance in situations like travelling around, performing any activity out of their daily habituated area like home (Shadi et al., 2019). The oldest paper in the research papers we reviewed goes back to 2009 offering a solution with image recognition technology with help of small mini cameras on the both ends of the glasses that person wears and the paper claims to develop a text reading system. From then the papers had solutions offered to people with low vision and visual impaired are using the following technologies Navbelt guidance system, Personal digital assistant, Inverse Dynamic Pattern System Algorithm, Color coding, Convolution Neural Networks and Recurrent Neural Networks, Text to speech engine, Deformable Grid, few geometric formulae and Algorithms, Technology used in autonomous cars is currently under testing for navigation of visually challenged people, Real time semantic segmentation, Radio Frequency Identification, ORB-SLAM2 is a versatile and accurate monocular SLAM system module, Various algorithms, ultrasonic sensors, text speech module, Tensor flow objection detection API and CNN (Meena et al., 2022; Mohbey et al., 2022) models and more (Balachandran et al., 2003). All the research papers their solutions, technologies and the limitations they have are mentioned in the paper chronologically.

EXISTING SOLUTIONS TO HELP VISUALLY IMPAIRED PEOPLE

Currently, there are few applications and also few devices which are helping visually impaired people to lead their life like others. But there are certain problems with the available applications and devices. Here are some of the applications and devices that are available:

VisualPal a Mobile App for Object Recognition for the Visually Impaired

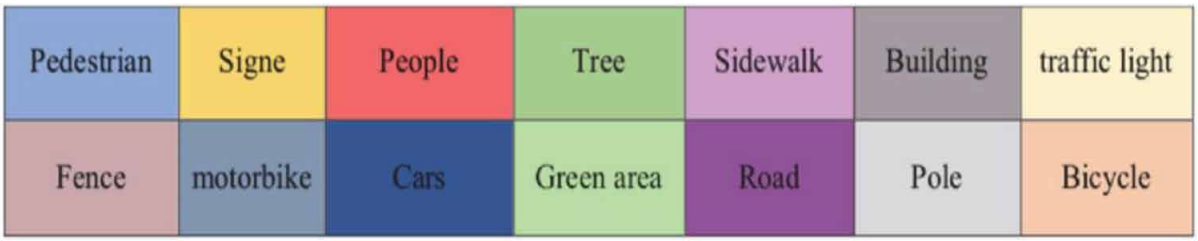
It is an application that is helping visually challenged users to recognize/identify objects. This application detects the color, name of the object that is given. Visual pal application uses Scale Invariant Feature Transform (SIFT) algorithm. This algorithm generally measures the Euclidean distance, if the variation is in range that is acceptable then the application spells the name/color of the image but when error is more than acceptable range then application classifies it as a new object and ask the user to enter the input which difficult for an impaired person. This application works as follows: When a visually impaired person captures an image, this application analyzes, processes the image and then gives a response in speech as output. But this application doesn't support recognition of landmarks, text and it also doesn't support voice input which is important for any application which is used by visually challenged people. Accuracy of this application registered around 71% and user interface is not much acceptable as visually impaired people find it difficult to use it. This application can be successful when it adds advanced technology and it should be cost effectively and user friendly (Bagwan & Sankpal, 2015).

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Outdoor Navigation for Visually Impaired based on Deep Learning

In this paper, the researchers proposed an outdoor navigation system based on deep learning. They mainly focused on providing real time applications that guide visually impaired people. Researchers added a new feature i.e., estimation of distance from person to object. This could help impaired people while walking in the outdoor environment.

Figure 1. Objects with different colors



This application has a dataset which is a key step in deep learning. The dataset is prepared based on data annotation, which means images of various objects are captured, collected and then labeled with a color shown in figure 1.

When a user of the application captures an image, it searches in the dataset based on the color coded to it. The training model used in this application is designed using DeepLabV3+, one the most powerful techniques which is used for semantic image segmentation in collaboration with deep learning (Hiran & Doshi, 2013). Application used some mathematics for estimation of distance between object and person. This application mainly guides the visually impaired people regarding obstacles while they are walking outdoors. The main problem with this application is the dataset is with a limited number of objects. The other problem is the distance calculation between object and the person is not accurate and has a high chance of error (Shadi et al., 2019).

Smart Guiding Glasses for Visually Impaired People in Indoor Environment

Researchers proposed glasses that are wearable for people who are in need of them (i.e., visually challenged) which can be used in an indoor environment. Main theme of developing the system is to avoid collisions between visually impaired people and other people. The proposed system will work as follows: Visually impaired people will wear a smart guiding glass which will scan the people and objects on their way and communicate them through audio output which can be heard using earphones. They tested various methods of communicating output but finally settled with beep sound when there is an obstacle in their way. The researchers used ultrasonic sensors, AR rendering and many others sensors in glasses. While testing they found that while using smart guiding glasses total collisions have decreased drastically. Also, the time for walking to a place has decreased by at least 10%. But the only problem with the proposed system is, it is only designed for indoor environments and it has failed in outdoor environments. That means the proposed system cannot guide impaired people while they travel outside. The proposed system overcame many problems of previously available systems but researchers should

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work to limit it, they should try to extend it to outdoor environments as well which seems to be tough ask for the researchers (Bai et al., 2017).

Visual Localizer Outdoor Localization Based on ConvNet Descriptor and Global Optimization for Visually Impaired Pedestrians

In this paper, researchers proposed a visual localizer. They have designed a wearable assistance system Intoer. This has inbuilt headphones, customized GNSS module and camera. Visually impaired people are guided in the outdoor environment. It acts as a smart GPS tracking device which navigates the people. The researchers created a database where millions of images were stored. The system checks the image and then sends the description of the image to the user within no time (Hiran & Doshi, 2013). When the proposed system is tested, error is in acceptable range. But image matching is done in offline mode. If they could do it on a real time basis, then the proposed system would succeed in helping visually impaired people (Vyas et al., 2021).

A Smart Personal AI Assistant for Visually Impaired People

In this paper, researchers proposed a system with advanced features. The proposed system can recognize objects, can analyze text and also can detect landmarks. They have designed a system that is very interactive with users. This proposed system overcame challenges of visualpal application. In Proposed AI assistant researchers used Google cloud speech API to give audio output and powerful algorithms to detect and process the image. The accuracy of the proposed smart personal AI was high and error was minimum (Hiran & Doshi, 2013). Proposed system also added a feature of currency recognition which is not available till date in applications helping visual impaired people. There are few drawbacks in this proposed system as well. Main drawback is this application doesn't support Multilingual which means it only works in only one language. One more drawback is it doesn't measure the distance between the object and person. Another drawback is that a visually challenged person always needs to use his mobile to scan the image/object instead researchers should have gone with providing a wearable device which is connected to a smartphone, which reduces the burden on visually impaired people (Felix et al., 2018), (Unogwu et al., 2022).

Design, Development and Clinical Evaluation of the Electronic Mobility Cane for Vision Rehabilitation

Researchers developed a system to help the people who are unable to detect the obstacle in their way. Proposed system uses a new electronic mobility cane (EMC). The main important feature of this technology is that it helps users to have a map of the surroundings in mind. Information of their surroundings in a simple way without overloading the information with the help of vibration, audio or through the voice output (Laxmi et al., 2021).

Proposed system has 6 ultrasonic sensors, which is helpful for the visually challenged people. However, there are challenges in the system, as the system needs to scan the area from the user's point of view all the time which can cause problems in the system. The canes which are used in the system use very narrow beam laser devices, which raises another limitation that it cannot detect the objects which are made up of glass as these laser pass through the transparent glasses easily, furthermore the cost of

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the system is slight high and it also requires user to have expertise to operate these devices. To conclude this system is suitable for an indoor environment with certain limitations and it is not at all feasible for an outdoor environment (Bhatlawande et al., 2014), (Meena et al., 2021).

Other Research Papers Offering Different Solutions to Visually Impaired People

Table 1. Important researches in area of solutions towards visually impaired people

S. No	Article	Technology Used	Proposed Solution	Limitations
1	(Balachandran et al., 2003)	System is designed using Inverse DPS Algorithm	GPS based Navigation System	Proposed system has low accuracy and it is using old technology
2	(Kumar & Meher, 2015)	Used Color coding, CNN and RNN technologies	Proposed system recognizes color and objects	System can detect very limited No.of objects
3	(Kang et al., 2017)	Deformable Grid, few geometric formulae and Algorithms	Provides enhanced obstacle avoidance	Sometimes proposed system fails to perform motion tracking
4	(Pham et al., 2016)	Ultrasonic Sensors, text speech module	Accurate prediction of distance	Not designed to detect colors and facial actions
5	(Sabab & Ashmafee, 2016)	Text to speech engine	Braille Note taker, Braille Printer	Obstacle detection is not available and only concentrated on reading.
6	(Hernández et al., 2009)	Optical Character Recognition software's, text reading software's	Concentrated on using sensory substitution approach	The system has many drawbacks in testing as researchers concluded that it requires more high end technology
7	(Farhath et al., 2014)	Database module, Label reading software's	Proposed a system can help visually challenged people in shopping	Accuracy in low as this application requires user to click whole picture of product
8	(Doshi et al., 2023)	Applied Deep Learning	Proposed a system to recognize handwritten text	Accuracy in recognizing complex images is very low
9	(Morrison et al., 2017)	3D printed tactile maps and AI tools	Researchers understood what visually people are expecting from AI based application	System designed should be cost effective and user friendly
10	(Svyrydov et al., 2018)	3 powerful algorithms and AI applications	Proposed a system to improve image processing	Proposed system just detects the object/image but does not guide the user.
11	(Samigulina et al., 2017)	Algorithm of the innovative technologies of DL PIV	Innovative method for providing distance learning to visually impaired people	Not preferred by users as the interface is not user friendly
12	(Manduchi, 2012)	ARM 11 332 MHz processor	Assist visually impaired people while they are walking	Only gives beep sound when there is obstacle which means there are no advanced features
13	(Real & Araujo, 2019)	ETA(Electronic travel aid) and LiDAR	Outdoor and Indoor navigation, smart vision	Smart phone is required all the times which could be a concern
14	(Jabnoun et al., 2014)	SIFT, SURF(Speeded up robust features)	Help in recognizing objects for visually impaired users	Restricted only to object recognition, needed advanced features
15	(Matusiak et al., 2013)	SIFT, DoG based algorithm	Recognizes objects for visually impaired users	Restricted only to object recognition, needed advanced features

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Table 1. Continued

S. No	Article	Technology Used	Proposed Solution	Limitations
16	(Saleous et al., 2016)	Rpi-based platform, Cloud OCR	Reads the data for the visually challenged users	Usability is difficulty for the users, not much user friendly
17	(Dasila et al., 2017)	Binaural audio module and Segmentation algorithms	Identification of objects with high accuracy	The testing system failed to detect small size items.
18	(Cheng et al., 2018)	Electronic Travel Aid and other powerful light detection algorithm	Providing assistance for visually impaired people while crossing roads	Narrow focus only on pedestrian who are crossing roads that has traffic lights
19	(Yang et al., 2018)	Real time semantic segmentation	Operates in real time, wearable, wireless & reliable device	Testing is done with limited No.of people so they need to test it in large numbers. There are certain trade offs made in efficiency.
20	(Martinez et al., 2017)	Technology used in autonomous cars is tested for guiding visually challenged people	Navigation of visually impaired people using a wearable device which has advanced technology	Still in the developing stage and only navigation is available, lacking features like distance and reader.
21	(Raghavan et al., 2021)	Tensor flow objection detection API and CNN models	Chabot and Real time Facial recognition	Person should have partial vision, application cannot calculate distance between objects and it's not wearable device
22	(Bai et al., 2018)	RFID, ORB-SLAM2 module, Various algorithms	Indoor and outdoor Navigation	Only navigation is available, lacking features like distance and reader.
23	(Dakopoulos & Bourbakis, 2009)	Navbelt guidance system, Personal digital assistant	Real time operations, Wireless Connectivity	Still in developing stage
24	(Sharmila et al., 2020)	IoT sensors, Principal Component Analysis Algorithm	Proposed system not only detects objects but also face recognition	Face recognition accuracy is not up to the mark and need improvement

RECOMMENDATIONS

1. The proposed system should be a wearable device but not using a smartphone.
2. The technology that is being developed should be cost effective.
3. Proposed system should have the following features: Object recognition, Distance calculation, Facial recognition, Text to speech, user friendly interface, navigation, GPS and also should support both indoor and outdoor environments.
4. Solutions that are offered cannot be solely used by visually impaired people, however they are helpful only as assistance to them. Such a point has to be considered in further solution developments.
5. Accuracy of image recognition has to be improved and suggested to be combined with machine learning technologies to make them more successful.
6. Daily life activities like payments and transactions have to make their solutions for visually challenged to be more authentic and easier to use.

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CONCLUDING REMARKS

There are numerous technologies that are developed over a period of time, though each of them is having their own limitations in aspect of usability, feasibility and even in cost to implement the solutions, many of the ideas were not that popular may be due to such limitations. Any small additional assistance to visually impaired and blind people brings a relief to their lives. In this paper we tried to bring the available papers with different technologies together at one place and show the limitations they have. Recommendations were given in the light of research papers reviewed.

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